

Admissibility Before Truth

On the Domain of Truth Predicates

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Abstract

Standard epistemology treats truth as the primary predicate and asks which beliefs satisfy it. This paper argues that truth is not primary but derived: it is a partial predicate whose domain is determined by a prior structure, the observer’s admissibility manifold. We prove the *Domain Restriction Theorem*: every well-defined truth predicate $\mathcal{T}$ factors through the inclusion $\iota : \text{Rest}(\mathcal{R}) \hookrightarrow \mathcal{D}$ of the restorable distinction space into the full distinction space, so that $\text{dom}(\mathcal{T}) \subseteq \text{Rest}(\mathcal{R})$. Distinctions outside the admissible set are not false; they are unevaluable, receiving the third value $\perp$ in a natural three-valued extension. The theorem is not a philosophical claim but a structural consequence of the repair-theoretic framework developed in the companion papers [1, 2]: truth-evaluation is itself a repair process, and can only operate where repair capacity exists. Consequences are derived for Bayesian inference, machine learning, legal admissibility, language acquisition, and the Kuhnian account of scientific revolutions. In each case, what had appeared to be a domain-specific epistemic constraint is revealed to be an instance of the same domain restriction. The paper is the third in a five-paper program; an appendix summarises the definitions required from the companion papers so that the central theorem can be read independently.

1. The Program and This Paper’s Place in It

This paper is the third in a sequence of five, each of which introduces one genuinely new layer of mathematical structure while relying on the previous layers rather than rebuilding them. The architecture is as follows.

The first paper, *Observability Is Restorability* [1], defines the primitive objects: repair operators, restorable distinctions, the admissibility manifold $A(\mathcal{R})$, and the repair pseudometric. Its central theorem is that the set of observable distinctions coincides with the set of restorable distinctions.

The second paper, *Erasure Is Deformation* [2], defines the dynamics of loss: repair-repertoire reduction deforms the admissibility manifold, generating metric singularities at the irrestorability boundary. Its central object is the pullback deformation tensor $E_t = l_t^* g_0 - g_t$ measuring how erasure distorts the geometry of what remains.

The present paper establishes the logical consequences: truth is a partial predicate whose domain is the admissible distinction space. The central theorem is the Domain Restriction Theorem (Theorem 4.1).

The fourth paper, *The Geometry of Witnesses*, will explain why admissibility persists over historical time, introducing second-order repair operators (witnesses that maintain repair operators rather than distinctions directly).

The fifth paper, *Conservation of Ambiguity*, will describe how finite repair resources are allocated across a living distinction space, with ambiguity, repair cost, and interference as competing claims on the same budget.

The present paper is designed to serve as a conceptual entry point to the program. Its central theorem is stated and proved using only the definitions in Section 2 and the Appendix; a reader encountering the program for the first time may read this paper first and return to the companion papers for full derivations.

Standard epistemology begins with truth and asks which beliefs satisfy it. This ordering is standard but not mandatory. The repair-theoretic account reverses it: an observer must first establish which distinctions it can maintain (admissibility), then identify which of those distinctions it can discriminate in context (the observable subset), and only then assign truth values. The ordering $A(\mathcal{R}) \rightarrow \mathcal{D}_O \rightarrow \mathcal{T}|_{\mathcal{D}_O}$ is not a philosophical preference. It is a theorem.

The paper proceeds as follows. Section 2 reviews the repair framework. Section 3 defines truth predicates formally. Section 4 states and proves the Domain Restriction Theorem. Section 5 demonstrates that a state-of-the-art hierarchical classification architecture (SAHC) independently implements the same structure, and derives the Hierarchy Consistency Corollary. Section ?? develops the three-valued semantics. Sections 6–10 derive consequences for Bayesian infer-

ence, machine learning, legal admissibility, language acquisition, and scientific revolutions. Section 11 shows the ordering does not reverse. Section 12 states open problems. Section 13 draws broader consequences. The Appendix collects required definitions from the companion papers.

2. Background: Repair, Distinctions, and Admissibility

We recall the definitions required from the companion papers. Full derivations appear in [1]; we state here only what is needed for the Domain Restriction Theorem.

A *substrate* $S = (X_S, \mathcal{D}, \mathcal{D})$ consists of a locally compact Hausdorff state space X_S , a perturbation semigroup $\mathcal{D}$, and a closed distinction relation $\mathcal{D} \subseteq X_S \times X_S$. A *repair repertoire* $\mathcal{R} \subseteq C(X_S, X_S)$ is a sequentially compact set of continuous maps. The *minimal repair cost* of distinction d under $\mathcal{R}$ is

$$C_R(d, \mathcal{R}) = \sup_{p \in \mathcal{D}_d} \inf_{\gamma \in \Gamma_d^p(\mathcal{R})} \text{cost}(\gamma),$$

setting $C_R(d, \mathcal{R}) = +\infty$ when no finite repair path exists. The *restorable distinction space* is

$$\text{Rest}(\mathcal{R}) = \{d \in \mathcal{D} \mid C_R(d, \mathcal{R}) < +\infty\}.$$

The *admissibility manifold* is

$$\mathcal{A}(\mathcal{R}) = \overline{\{x \in X_S \mid \forall d \in \text{Rest}(\mathcal{R}), C_R(d, \mathcal{R}) < +\infty \text{ from } x\}}.$$

The Repair Boundary Theorem [1] states that $x \in \partial \mathcal{A}(\mathcal{R})$ if and only if $C_R(d, \mathcal{R}) \rightarrow +\infty$ for some $d \in \text{Rest}(\mathcal{R})$ as the system approaches x .

Remark 2.1. An observer is, in the repair-theoretic account, a pair $(S, \mathcal{R})$. What the observer can observe is exactly $\text{Rest}(\mathcal{R})$. What the observer cannot observe is not hidden from it; it is simply not part of the observer's world. The admissibility manifold $\mathcal{A}(\mathcal{R})$ is the geometric structure of that world.

3. Truth as a Partial Predicate

The classical notion of a truth predicate takes the form $\mathcal{T} : \mathcal{D} \rightarrow \{0, 1\}$, assigning a truth value to every distinction in the full distinction space. We argue that this is the wrong domain.

Definition 3.1 (Truth predicate). A *truth predicate* for observer $(S, \mathcal{R})$ is a function $\mathcal{T} : X \rightarrow \{0, 1, \perp\}$ defined on some subset $X \subseteq \mathcal{D}$, where 0 denotes falsity, 1 denotes truth, and $\perp$ denotes unevaluability. The predicate is *total* if $X = \mathcal{D}$; it is *partial* if $X \subsetneq \mathcal{D}$.

Definition 3.2 (Evaluability). A distinction $d \in \mathcal{D}$ is *evaluable* by observer $(S, \mathcal{R})$ if assigning a truth value to d is well-defined, i.e., if $d \in \text{dom}(\mathcal{T})$ with $\mathcal{T}(d) \in \{0, 1\}$. A distinction is *unevaluable* if $\mathcal{T}(d) = \perp$.

The classical picture assumes total truth predicates. We show this assumption is incoherent for finite observers.

Lemma 3.3 (Truth-evaluation is repair). *Any procedure for evaluating the truth of distinction d must be capable of discriminating d from its negation after arbitrary admissible perturbation. Formally, truth-evaluation of d requires the existence of a finite-cost repair path restoring d after perturbation. Therefore truth-evaluation of d presupposes $d \in \text{Rest}(\mathcal{R})$.*

Proof. Evaluating whether d holds requires the evaluator to be in a state from which d is discriminable: the evaluator must be able to distinguish the case $(x, y) \in \mathcal{D}$ from the case $(x, y) \notin \mathcal{D}$. If d has been erased by an admissible perturbation p , the evaluator must restore discriminability before it can evaluate. The restoration of discriminability after perturbation is precisely the repair of d : a finite-cost repair path $\gamma \in \Gamma_d^p(\mathcal{R})$ returning the system to a state where d is discriminable. If no such path exists — if $C_{\mathcal{R}}(d, \mathcal{R}) = +\infty$ — then no evaluation procedure can operate, because there is no state the evaluator can reach from which the discrimination can be made. Hence $d \notin \text{Rest}(\mathcal{R})$ implies d is unevaluable. $\square$

Remark 3.4. Lemma 3.3 does not claim that truth-evaluation literally requires executing a repair sequence. It claims that the *capacity* to evaluate truth requires the *existence* of such a sequence. An observer that cannot in principle restore d

cannot in principle evaluate d , because it cannot reliably distinguish the truth condition from states reached by perturbation. This is an operational constraint, not a claim about the phenomenology of evaluation.

4. The Domain Restriction Theorem

Theorem 4.1 (Domain Restriction Theorem). *Let $(S, \mathcal{R})$ be an observer and let $\mathcal{I}$ be a well-defined truth predicate for this observer. Then $\mathcal{I}$ factors through the inclusion $\iota : \text{Rest}(\mathcal{R}) \hookrightarrow \mathcal{D}$:*

$$\mathcal{I} = \tilde{\mathcal{I}} \circ \iota,$$

where $\tilde{\mathcal{I}} : \text{Rest}(\mathcal{R}) \rightarrow \{0, 1\}$ is the restriction of $\mathcal{I}$ to the restorable distinction space. Consequently,

$$\text{dom}(\mathcal{I}) \subseteq \text{Rest}(\mathcal{R}).$$

A well-defined truth predicate cannot assign truth values outside the admissible distinction space.

Proof. By Lemma 3.3, every distinction in $\text{dom}(\mathcal{I})$ must be in $\text{Rest}(\mathcal{R})$. Therefore $\text{dom}(\mathcal{I}) \subseteq \text{Rest}(\mathcal{R})$.

Define $\tilde{\mathcal{I}} : \text{Rest}(\mathcal{R}) \rightarrow \{0, 1\}$ by $\tilde{\mathcal{I}}(d) = \mathcal{I}(\iota(d)) = \mathcal{I}(d)$ for all $d \in \text{Rest}(\mathcal{R})$. Since ι is the inclusion map (the identity on its domain), $\mathcal{I} = \tilde{\mathcal{I}} \circ \iota$ on $\text{dom}(\mathcal{I})$. For $d \notin \text{Rest}(\mathcal{R})$, $\mathcal{I}(d) = \perp$ by definition of unevaluability, which is consistent with $\tilde{\mathcal{I}}$ being undefined outside $\text{Rest}(\mathcal{R})$. $\square$

Remark 4.2 (The categorical reading). The factorization $\mathcal{I} = \tilde{\mathcal{I}} \circ \iota$ is a universal property: every truth predicate factors through ι . In categorical terms, $\iota : \text{Rest}(\mathcal{R}) \hookrightarrow \mathcal{D}$ is the unique morphism through which all well-defined truth predicates factor. This means that the inclusion of the admissible distinction space into the full distinction space is the *domain object* for truth predicates: any construction involving truth automatically inherits the admissibility constraint via this factorization.

Corollary 4.3 (The strict ordering). *The ordering*

$$A(\mathcal{R}) \longrightarrow \text{Rest}(\mathcal{R}) \longrightarrow \mathcal{I}|_{\text{Rest}(\mathcal{R})}$$

is strict at each arrow. Admissibility determines the observer's world; restorability determines the domain of truth; truth is a predicate on that domain. No step in this sequence can be reversed or collapsed without contradiction.

Proof. The first arrow is strict because $\mathcal{A}(\mathcal{R})$ is a subset of X_S , not of $\mathcal{D}$; the map sends the state manifold to the distinction space via the induced discrimination structure. The second arrow is strict because $\text{Rest}(\mathcal{R}) \subsetneq \mathcal{D}$ in general: there exist distinctions in $\mathcal{D}$ that are not restorable. The third arrow is strict because $\mathcal{T}$ is a partial function on $\text{Rest}(\mathcal{R})$, not all of $\text{Rest}(\mathcal{R})$ is evaluable in practice even if it is admissible in principle. $\square$

Theorem 4.4 (Geometry of the evaluability boundary). *The boundary of $\text{dom}(\mathcal{T})$ within $\mathcal{D}$ coincides with the irrestorability boundary: the set of distinctions d for which $C_R(d, \mathcal{R}) \rightarrow +\infty$. The domain of truth therefore inherits the geometric structure of the admissibility manifold: it has a well-defined metric (the repair pseudometric), a boundary (where repair cost diverges), and topological invariants (the homology groups of the admissibility manifold).*

Proof. $\text{dom}(\mathcal{T}) \subseteq \text{Rest}(\mathcal{R})$ by the Domain Restriction Theorem. The boundary of $\text{Rest}(\mathcal{R})$ within $\mathcal{D}$ is exactly the set of distinctions approaching which $C_R(d, \mathcal{R}) \rightarrow +\infty$, by the Repair Boundary Theorem of [1]. The repair pseudometric d_R on $\text{Rest}(\mathcal{R})$ and the homology groups $H_*(\mathcal{A}(\mathcal{R}))$ are defined in [1, 2] and are inherited by $\text{dom}(\mathcal{T})$ via the inclusion $\text{dom}(\mathcal{T}) \subseteq \text{Rest}(\mathcal{R})$. $\square$

Remark 4.5. Theorem 4.4 means that evaluability is not a flat binary partition of all possible claims into evaluable and unevaluable. The domain of truth has boundary structure: some distinctions are marginally evaluable (near the irrestorability boundary, with high repair cost), and the geometry of that margin is non-trivial. Near-boundary evaluability corresponds to claims that can barely be assessed, that require enormous epistemic effort, or that become unevaluable under small perturbation. This is the formal correlate of the philosophical notion of “borderline cases.”

5. Hierarchical Domain Restriction: Empirical Instantiation

The Domain Restriction Theorem is a mathematical result. This section argues that it is also an empirical one: a state-of-the-art deep learning architecture for

satellite image classification independently converged on the same structure, and its performance gains over non-hierarchical baselines constitute evidence that the domain restriction is a genuine design principle, not a philosophical preference.

5.1. The Architecture as Factorization

Weikmann et al. [3] propose the Semantics-Aware Hierarchical Consensus (SAHC) method for multi-level remote sensing classification. Their architecture introduces H classification heads $g^h \circ f$, one per hierarchical level, together with trainable hierarchy log-joint matrices $J^{h,h'}$ that project predictions between levels, and a consensus loss L_{HC} penalising disagreement among all projected predictions at a target level. The headline result is consistent improvement over non-hierarchical baselines across every level of a 90-class taxonomy [3].

We claim that this architecture is an empirical instance of the Domain Restriction Theorem applied at multiple levels simultaneously. Let $h = 1$ (coarse), $h = 2$ (intermediate), $h = 3$ (fine) denote the three classification granularities. Define:

- $\mathcal{R}_1$: the repair repertoire sufficient to restore coarse-level distinctions (vegetation vs. non-vegetation; urban vs. rural);
- $\mathcal{R}_2 \supseteq \mathcal{R}_1$: the repair repertoire sufficient to restore intermediate distinctions (forest vs. shrub; residential vs. industrial);
- $\mathcal{R}_3 \supseteq \mathcal{R}_2$: the repair repertoire sufficient to restore fine-grained distinctions (deciduous forest vs. coniferous; compact residential vs. sparse residential).

By Repair Monotonicity [1], $\text{Rest}(\mathcal{R}_1) \subseteq \text{Rest}(\mathcal{R}_2) \subseteq \text{Rest}(\mathcal{R}_3)$, so the admissibility manifolds are nested: $\mathcal{A}(\mathcal{R}_1) \subseteq \mathcal{A}(\mathcal{R}_2) \subseteq \mathcal{A}(\mathcal{R}_3)$. The Domain Restriction Theorem then yields a chain of inclusions for the truth predicates at each level:

$$\mathcal{T}_f = \tilde{\mathcal{T}}_f \circ \iota_{f,m} \circ \iota_{m,c},$$

where $\iota_{m,c} : \text{Rest}(\mathcal{R}_2) \hookrightarrow \text{Rest}(\mathcal{R}_1)$ and $\iota_{f,m} : \text{Rest}(\mathcal{R}_3) \hookrightarrow \text{Rest}(\mathcal{R}_2)$ are the admissibility-preserving inclusions. This says: the fine-level truth predicate $\mathcal{T}_f$

factors through the intermediate admissibility domain, which in turn factors through the coarse admissibility domain. A pixel cannot be truthfully classified as “deciduous forest” unless it is first admissible as “forest” and before that as “vegetation.”

5.2. Hierarchy Matrices as Approximate Inclusion Maps

SAHC’s trainable matrices $J^{h,h'} \in [0,1]^{|\Omega^h| \times |\Omega^{h'}|}$ are initialised from a user-defined binary hierarchy and then learned from data. In the repair-theoretic reading, $J^{h,h'}$ is an empirical approximation of $\iota_{h,h'}$: it encodes the learned conditional probability that a fine-level class ω_i^h lies within the admissibility domain of a coarser class $\omega_j^{h'}$. Where the user-defined hierarchy is rigid, $J^{h,h'}$ learns which semantic overlaps the rigid taxonomy missed.

The heatmap of the learned joint probability matrix J^{h_2,h_1} on the ELU dataset (reported in [3]) confirms this interpretation: the highest-weight entries align with the predefined hierarchy, but off-diagonal weights reveal genuine semantic adjacencies — “water basins” and “aquaculture in the marine environment,” “continuous urban fabric” and “road networks” — that the rigid taxonomy excluded but the data supports. These off-diagonal weights are not noise. They are the learned repair cost relationships: two classes are semantically adjacent precisely when the repair operators for one partially restore the other.

5.3. The Consensus Loss as Domain Restriction Penalty

SAHC’s hierarchical consensus loss L_{HC} penalises the Jensen–Shannon divergence between the consensus prediction $\hat{p}_{HC}^{\tilde{h}}$ and the projected predictions from every other level:

$$L_{HC} = \frac{1}{N} \sum_n \sum_{\tilde{h}} \frac{1}{\log |\Omega^{\tilde{h}}|} \sum_h \text{JSD}^{h,\tilde{h}}(\hat{p}_{HC}^{\tilde{h}} \| \hat{p}^{h \rightarrow \tilde{h}}).$$

In the repair-theoretic reading, this is the penalty for violating the Domain Restriction Theorem:

$$L_{HC} \approx d(\tilde{\mathcal{I}} \circ \iota, \mathcal{I}),$$

where d is a divergence between the admissibility-restricted prediction and the unconstrained prediction. When $L_{HC} = 0$, the classifier’s fine-level outputs are

exactly consistent with their coarse-level projections: fine distinctions are admissible at every level above them. When $L_{HC} > 0$, the classifier has assigned a truth value to a fine distinction that cannot be projected into the coarser admissibility domains — it has predicted “deciduous forest” where the coarse head predicts “urban area.” SAHC penalises this prediction as inadmissible.

The ablation study in [3] isolates this penalty’s contribution: adding L_{HC} to the multi-head baseline raises fine-grained OA from 96.44% to 96.52%, and adding both consensus training and consensus inference raises it further to 96.63%. The performance gains are not large in absolute terms, but their mechanism is precisely the domain restriction: forcing fine-level truth assignments to be restorable through coarser admissibility levels reduces inadmissible predictions and improves accuracy.

5.4. Generalisation: From SAHC to TARTAN

SAHC assumes a fixed semantic hierarchy: levels are defined by a human-specified taxonomy, and hierarchy matrices encode transitions between taxonomy levels. This is one concrete instance of a more general structure.

In the TARTAN framework, reality is represented as a hierarchy of interacting tiles at arbitrary resolutions. Each level carries its own admissibility constraints, and cross-scale consistency is maintained through transport operators $T_{i \rightarrow j}$ that need not follow a fixed semantic taxonomy. The TARTAN coherence functional,

$$L_{\text{TARTAN}} = \sum_{i < j} d(T_{i \rightarrow j}(x_i), x_j),$$

enforces the Domain Restriction Theorem simultaneously across spatial, temporal, semantic, and computational scales. SAHC’s $J^{h,h'}$ matrices are one specific parameterisation of $T_{i \rightarrow j}$ in the semantic hierarchy case; SAHC’s L_{HC} is one specific instance of L_{TARTAN} under Jensen–Shannon divergence.

The formal correspondence is:

Hierarchy levels $h \in [1, H]$	$\leftrightarrow$	Resolution levels $i \in I$
Hierarchy matrix $J^{h,h'}$	$\leftrightarrow$	Transport operator $T_{i \rightarrow j}$
Projected prediction $\hat{p}^{h \rightarrow h'}$	$\leftrightarrow$	Transported state $T_{i \rightarrow j}(x_i)$
Consensus loss L_{HC}	$\leftrightarrow$	Coherence functional L_{TARTAN}
Semantic taxonomy	$\leftrightarrow$	Arbitrary resolution hierarchy

SAHC is therefore not merely a hierarchical classifier but a specific instance of admissibility-preserving multiscale computation. Its empirical success across two datasets with very different hierarchical complexity suggests that the Domain Restriction Theorem is not only a logical consequence of the repair framework but a productive engineering principle when implemented at scale.

5.5. A Formal Consequence

Corollary 5.1 (Hierarchy Consistency Corollary). *Let $\mathcal{R}_1 \subseteq \mathcal{R}_2 \subseteq \dots \subseteq \mathcal{R}_H$ be a hierarchy of repair repertoires with admissibility-preserving inclusions $\iota_{h,h'} : \text{Rest}(\mathcal{R}_h) \hookrightarrow \text{Rest}(\mathcal{R}_{h'})$ for all $h > h'$. Suppose a prediction architecture produces truth predicates $\mathcal{T}_h$ at each level satisfying*

$$\mathcal{T}_h = \tilde{\mathcal{T}}_h \circ \iota_{h,h'} \quad \text{for all } h > h'.$$

Then each $\mathcal{T}_h$ satisfies the Domain Restriction Theorem at its own level, and the composition $\mathcal{T}_H = \tilde{\mathcal{T}}_H \circ \iota_{H,H-1} \circ \dots \circ \iota_{2,1}$ satisfies it simultaneously at all levels.

Proof. By the Domain Restriction Theorem applied to each level: $\text{dom}(\mathcal{T}_h) \subseteq \text{Rest}(\mathcal{R}_h)$ for each h . Since $\mathcal{T}_h = \tilde{\mathcal{T}}_h \circ \iota_{h,h'}$ and $\iota_{h,h'}$ maps $\text{Rest}(\mathcal{R}_h)$ into $\text{Rest}(\mathcal{R}_{h'})$, evaluability at level h implies evaluability at level h' . The composed factorization follows by induction on the number of levels. $\square$

Remark 5.2. Corollary 5.1 is the formal statement that SAHC-style architectures, when their hierarchy matrices exactly implement the admissibility-preserving inclusions $\iota_{h,h'}$, simultaneously satisfy the Domain Restriction Theorem at every level. SAHC’s learned matrices $J^{h,h'}$ are approximate implementations of $\iota_{h,h'}$; the consensus loss L_{HC} is the gradient signal that drives them toward exactness. The residual violation — $L_{HC} > 0$ at convergence — measures the extent to which the architecture departs from strict multi-level domain restriction.

Architectures with smaller L_{HC} at convergence are, in this precise sense, better implementations of the theorem.

The Domain Restriction Theorem introduces $\perp$ as a third truth value for unevaluable distinctions. This is not a mere convenience; the three-valued structure follows necessarily from the repair framework.

Proposition 5.3 (Unevaluability is not falsity). *If $d \notin \text{Rest}(\mathcal{R})$, then $\mathcal{T}(d) \neq 0$. An unevaluable distinction is not false.*

Proof. Assigning $\mathcal{T}(d) = 0$ (falsity) would itself be a truth assignment, which by the Domain Restriction Theorem requires $d \in \text{dom}(\mathcal{T}) \subseteq \text{Rest}(\mathcal{R})$. But we assumed $d \notin \text{Rest}(\mathcal{R})$. Contradiction. Therefore $\mathcal{T}(d) \neq 0$. By the same argument, $\mathcal{T}(d) \neq 1$. Hence $\mathcal{T}(d) = \perp$. $\square$

Remark 5.4. Proposition 5.3 resolves a persistent confusion in accounts of semantic unevaluability. Unevaluable distinctions are sometimes treated as “false by default” (no truth-maker, no truth) or as “indeterminate” (neither true nor false). The repair-theoretic account gives a precise meaning to neither: $\perp$ is not a weakened form of falsity or truth but a distinct value arising from the failure of the precondition for evaluation. The natural semantic framework is Kleene’s strong three-valued logic [5], in which $\perp$ propagates through connectives in a specific way. Whether the repair-theoretic $\perp$ satisfies the Kleene truth tables exactly is a question for the logic paper this framework suggests; we leave it as an open problem.

Proposition 5.5 (Admissibility expansion increases evaluability). *If $\mathcal{R}_1 \subseteq \mathcal{R}_2$, then $\text{dom}(\mathcal{T}^{(1)}) \subseteq \text{dom}(\mathcal{T}^{(2)})$. Acquiring more repair operators strictly expands the domain of truth.*

Proof. By the Repair Monotonicity Theorem of [1], $\mathcal{R}_1 \subseteq \mathcal{R}_2$ implies $\text{Rest}(\mathcal{R}_1) \subseteq \text{Rest}(\mathcal{R}_2)$. Since $\text{dom}(\mathcal{T}^{(i)}) \subseteq \text{Rest}(\mathcal{R}_i)$, the inclusion follows. $\square$

Remark 5.6. Proposition 5.5 is the logical analogue of the “Learning enlarges worlds” corollary from [1]. There it meant that acquiring repair operators expands the observer’s perceptual world. Here it means acquiring repair operators expands the domain of truth: claims that were unevaluable become evaluable. Learning is not merely acquiring new true beliefs; it is expanding the space of evaluable claims.

6. Bayesian Inference

Standard Bayesian inference assigns a prior $P : \mathcal{D} \rightarrow [0, 1]$ to every distinction in the full distinction space, with $P(d)$ interpreted as the probability that d is true. The Domain Restriction Theorem imposes a constraint on this picture that has not previously been made explicit.

Proposition 6.1 (Coherence requires admissibility). *A Bayesian prior P is coherent for observer $(S, \mathcal{R})$ only if $\text{supp}(P) \subseteq \text{Rest}(\mathcal{R})$. Assigning a non-trivial prior to an inadmissible distinction is incoherent.*

Proof. A non-trivial prior $P(d) \in (0, 1)$ presupposes that d can in principle be evaluated: otherwise the probability cannot be updated by evidence. But by the Domain Restriction Theorem, d is evaluable only if $d \in \text{Rest}(\mathcal{R})$. Therefore $P(d) \notin \{0, 1\}$ requires $d \in \text{Rest}(\mathcal{R})$, and the support of any coherent prior lies within $\text{Rest}(\mathcal{R})$. $\square$

Remark 6.2. Proposition 6.1 does not prohibit assigning $P(d) = 0$ to inadmissible distinctions; it prohibits treating that assignment as meaningful. A prior that places zero probability on $d \notin \text{Rest}(\mathcal{R})$ is technically coherent but trivially so: it says nothing about d because d cannot be evaluated. The operationally meaningful content of a Bayesian prior is entirely concentrated on $\text{Rest}(\mathcal{R})$.

This has a practical consequence: bounded rationality is not a cognitive limitation requiring separate modelling. It is the necessary consequence of finite $\text{Rest}(\mathcal{R})$. A perfectly rational Bayesian agent with finite repair capacity has a prior supported on a proper subset of all logically possible distinctions. The shape of that subset is determined by the agent’s admissibility manifold, not by computational constraints.

7. Machine Learning

A classifier trained to distinguish class A from class B is, in repair-theoretic terms, constructing an approximation to a repair operator for the distinction $d = (A, B)$. Training is the process of building $\mathcal{R}$ such that $d \in \text{Rest}(\mathcal{R})$.

Proposition 7.1 (Out-of-distribution failure is inadmissibility). *Let d_{train} be the distinction for which a classifier was trained and let d_{deploy} be the distinction it faces*

at deployment. Out-of-distribution failure occurs when $d_{\text{deploy}} \notin \text{Rest}(\mathcal{R}_{\text{train}})$: the distinction to be evaluated was not in the admissibility manifold of the training repair repertoire.

Proof. Training builds repair operators sufficient to restore d_{train} after the perturbations seen during training. At deployment, if d_{deploy} differs from d_{train} in ways that require repair operators not in $\mathcal{R}_{\text{train}}$, then $C_R(d_{\text{deploy}}, \mathcal{R}_{\text{train}}) = +\infty$ and the classifier cannot evaluate d_{deploy} . This is the formal content of out-of-distribution failure. $\square$

Remark 7.2. The standard account of distribution shift appeals to statistical distance between training and deployment distributions. The repair-theoretic account gives a geometric alternative: the relevant quantity is whether the deployment distinction lies within or outside the admissibility manifold of the trained repair repertoire. These accounts are not equivalent. Two distributions can be statistically close while the deployment distinction lies outside the training admissibility manifold (if the distributions agree on most features but differ on the specific repair-relevant features). Conversely, distributions can be statistically distant while the deployment distinction remains admissible. The geometric account is more fundamental.

8. Legal Admissibility

Legal systems provide the most explicit real-world instance of the domain restriction structure. Rules of evidence define which distinctions are admissible for truth-evaluation in legal proceedings. The phrase “admissibility of evidence” is not a metaphor for the formal structure developed here; it is the same structure at the institutional level.

Definition 8.1 (Legal admissibility manifold). The *legal admissibility manifold* of a legal system is $\mathcal{A}(\mathcal{R}_{\text{legal}})$, where $\mathcal{R}_{\text{legal}}$ consists of the evidentiary and procedural rules that constitute the system’s repair repertoire. A distinction d between legal and illegal conduct is evaluable by the court if and only if $d \in \text{Rest}(\mathcal{R}_{\text{legal}})$.

Proposition 8.2 (Verdicts are domain-restricted). A legal verdict assigns a truth value (1 = guilty, 0 = not guilty) to a distinction d . By the Domain Restriction Theorem, this is only coherent if $d \in \text{Rest}(\mathcal{R}_{\text{legal}})$. Evidence that falls outside the legal

admissibility manifold does not make a verdict false; it makes the verdict unevaluable through that evidence.

Remark 8.3. This accounts for two features of legal reasoning that seem paradoxical from a purely truth-theoretic perspective. First, an inadmissible piece of evidence is not excluded because it is false or unreliable; it is excluded because the legal system’s repair repertoire cannot incorporate it. The distinction it would resolve is not within $\text{Rest}(\mathcal{R}_{\text{legal}})$. Second, a verdict of “not guilty” is not a verdict of “innocent”; it is a verdict that the distinction between guilt and innocence was not established within the admissibility manifold of the proceeding. Both features follow immediately from the domain restriction.

9. Language Acquisition

A child acquiring a language is expanding $\text{Rest}(\mathcal{R})$ in a specific domain: the phonological, syntactic, and semantic distinctions that the language encodes.

Proposition 9.1 (Grammaticality is admissibility). *A sentence s is grammatical in language L if and only if the distinctions s expresses are in $\text{Rest}(\mathcal{R}_L)$, where $\mathcal{R}_L$ is the repair repertoire associated with competence in L . An ungrammatical sentence does not express a false proposition; it fails to express a proposition at all, receiving $\mathcal{I}(s) = \perp$.*

Proof. Grammaticality is the condition under which a sentence picks out a recoverable distinction in the language’s semantic space. An ungrammatical sentence is one for which no repair path exists from the noisy or degraded signal to a well-formed distinction: $C_R(d_s, \mathcal{R}_L) = +\infty$. By the Domain Restriction Theorem, $\mathcal{I}(d_s) = \perp$. $\square$

Remark 9.2. Proposition 9.1 makes precise a claim that linguists and philosophers have approached informally: that truth-conditional semantics presupposes grammaticality. A sentence must be grammatical before it can be true or false. The repair-theoretic account gives this presupposition a formal home: grammaticality is admissibility, and admissibility is prior to truth by the domain restriction.

Language acquisition is repair-repertoire expansion. A child who learns a new grammatical construction acquires a new class of repair operators, expanding $\text{Rest}(\mathcal{R})$ and therefore expanding $\text{dom}(\mathcal{I})$. The child gains not merely new true beliefs but the capacity to evaluate a larger class of claims.

10. Kuhn Without Relativism

The Kuhnian account of scientific revolutions holds that scientists in different paradigms inhabit different conceptual worlds and that their claims are incommensurable. Critics have argued that this collapses into relativism: if there is no shared domain of evaluation, there is no fact of the matter about which paradigm is better.

The repair-theoretic account dissolves this debate by making the structure precise.

Theorem 10.1 (Paradigm shifts as domain transitions). *Let $\mathcal{R}_{old}$ and $\mathcal{R}_{new}$ be the repair repertoires of two paradigms. A non-trivial paradigm shift satisfies*

$$\text{Rest}(\mathcal{R}_{old}) \triangle \text{Rest}(\mathcal{R}_{new}) \neq \emptyset,$$

where $\triangle$ denotes symmetric difference. By the Domain Restriction Theorem, the domains of truth differ:

$$\text{dom}(\mathcal{T}^{old}) \triangle \text{dom}(\mathcal{T}^{new}) \neq \emptyset.$$

Some previously evaluable claims become unevaluable; some previously unevaluable claims become evaluable.

Proof. The first statement is the Paradigm Shift Theorem proved in [1]. The second follows immediately by applying the Domain Restriction Theorem to both repertoires. $\square$

Remark 10.2. Theorem 10.1 gives precise content to Kuhnian incommensurability without relativism. Incommensurability is not the claim that two paradigms disagree about the same facts. It is the structural claim that their domains of truth differ: $\text{dom}(\mathcal{T}^{old}) \triangle \text{dom}(\mathcal{T}^{new}) \neq \emptyset$. Claims in the symmetric difference are evaluable in one paradigm and unevaluable in the other.

This is not relativism because the structure is observer-relative, not truth-relative. There is no claim that the same distinction is both true and false. There is only the structural fact that different repair repertoires generate different domains of evaluability. Whether one paradigm is better than another is a question about which admissibility manifold is more productive, more robust, or more adequate to the phenomena — a comparison that is perfectly meaningful without requiring a view from nowhere.

11. The Limits of the Ordering

The Domain Restriction Theorem establishes that admissibility precedes truth. A natural question is whether this ordering can be reversed: can truth-evaluation expand the admissibility manifold?

Proposition 11.1 (The ordering does not reverse). *Truth-evaluation cannot expand $\text{Rest}(\mathcal{R})$. Evaluating the truth of $d \in \text{Rest}(\mathcal{R})$ does not add any distinction to $\text{Rest}(\mathcal{R})$.*

Proof. Truth-evaluation operates on distinctions already in $\text{Rest}(\mathcal{R})$: it assigns 0 or 1 to d . It does not create new repair operators. Therefore $\text{Rest}(\mathcal{R})$ is unchanged by truth-evaluation. Only acquiring new repair operators (expanding $\mathcal{R}$) can expand $\text{Rest}(\mathcal{R})$, and that is a separate act from evaluation. $\square$

Remark 11.2. The ordering $\mathcal{A}(\mathcal{R}) \rightarrow \text{Rest}(\mathcal{R}) \rightarrow \mathcal{T}|_{\text{Rest}(\mathcal{R})}$ therefore forms a strict hierarchy with no feedback from truth to admissibility. Feedback does exist in practice: discovering that a distinction is evaluable and important can motivate acquiring new repair operators, thereby expanding admissibility. But this feedback passes through the agent's actions, not through the logical structure of truth-evaluation itself. The logical ordering is irreversible; the causal story is richer.

12. Open Problems

1. **Three-valued semantics.** Does the repair-theoretic $\perp$ satisfy Kleene's strong three-valued truth tables exactly, or does it require a different connective structure? The propagation of $\perp$ through conjunction, disjunction, and negation in the repair framework may differ from Kleene's tables in cases where one conjunct is evaluable and the other is not. A formal comparison with paraconsistent and partial-logic frameworks [5] is needed.
2. **The topology of evaluability.** Theorem 4.4 establishes that $\text{dom}(\mathcal{T})$ inherits the topology of $\mathcal{A}(\mathcal{R})$. What are the topological invariants of the evaluability domain for various natural observers (logical agents, scientific communities, legal systems)? Is the evaluability domain always connected? What is the significance of non-trivial $H_1(\text{dom}(\mathcal{T}))$?

3. **Degrees of evaluability.** The current framework is binary: a distinction is either evaluable or not. A graded generalisation would define evaluability as a function of $C_R(d, \mathcal{R})$: high repair cost corresponds to marginal evaluability, low repair cost to robust evaluability. This would give a natural account of “hard cases” in law, “borderline cases” in vagueness, and “underdetermined” hypotheses in philosophy of science.
4. **Joint evaluability and interference.** When is a set of distinctions jointly evaluable? The interference matrix from [1] measures repair conflict between distinctions. High interference ($I(d_1, d_2) > 0$) means jointly evaluating d_1 and d_2 costs more than evaluating each separately. Is there a notion of a *consistent evaluation domain*: a maximal set of distinctions that can be jointly evaluated within a given repair budget?
5. **Truth predicates across paradigms.** Theorem 10.1 identifies the symmetric difference of evaluability domains across a paradigm shift. Can the repair pseudometric be used to define a distance between paradigms: how much does the evaluability domain change? A small symmetric difference with low repair cost corresponds to a minor revision; a large or topologically complex one corresponds to a major revolution.

13. Consequences

Taking the Domain Restriction Theorem seriously reorganises a range of standing debates.

Epistemology. The classical problem of the criterion — how do we know what we know without already knowing how to know it — is dissolved by the domain restriction. The criterion is not epistemic but operational: the admissibility manifold $A(\mathcal{R})$ is determined by the observer’s repair capacity, not by a prior epistemic standard. The regress terminates not in a foundational truth but in a repair repertoire.

Philosophy of science. Scientific progress is not merely the accumulation of true beliefs. It is the expansion of the domain of evaluability: acquiring repair operators that make previously unevaluable distinctions evaluable. The history of science is the history of $\text{Rest}(\mathcal{R})$ growing.

Philosophy of language. Meaning is not prior to truth; admissibility is prior to both. A meaningful sentence is one that picks out an admissible distinction; a true sentence is one that correctly characterises that distinction. The admissibility manifold is the formal home for what Wittgenstein called the limits of one's world.

Formal logic. Classical two-valued logic is the limiting case in which the observer's repair repertoire is unlimited: $\text{Rest}(\mathcal{R}) = \mathcal{D}$. Three-valued and paraconsistent logics are not alternatives to classical logic but natural consequences of finite repair capacity. The logical framework appropriate to a finite observer is determined by its admissibility manifold.

Admissibility is not a property truth has or lacks. It is the precondition for truth having anything to say.

A. Summary of Required Definitions

This appendix collects the definitions from [1] required for the main results. It is included so that the Domain Restriction Theorem (Theorem 4.1) can be read independently. Full derivations, axioms, and proofs appear in [1].

A.1 Substrate. A *substrate* $S = (X_S, \mathcal{P}, \mathcal{D})$ consists of a locally compact Hausdorff state space X_S , a perturbation semigroup $\mathcal{P}$, and a closed distinction relation $\mathcal{D} \subseteq X_S \times X_S$.

A.2 Repair repertoire. A *repair repertoire* $\mathcal{R} \subseteq C(X_S, X_S)$ is a sequentially compact set of continuous maps (repair operators) under the compact-open topology.

A.3 Minimal repair cost. $C_R(d, \mathcal{R}) = \sup_{p \in \mathcal{P}_d} \inf_{\gamma \in \Gamma_d^p(\mathcal{R})} \text{cost}(\gamma)$, with $C_R(d, \mathcal{R}) = +\infty$ if no finite repair path exists.

A.4 Restorable distinction space. $\text{Rest}(\mathcal{R}) = \{d \in \mathcal{D} \mid C_R(d, \mathcal{R}) < +\infty\}$.

A.5 Admissibility manifold. $\mathcal{A}(\mathcal{R}) = \overline{\{x \in X_S \mid \forall d \in \text{Rest}(\mathcal{R}), C_R(d, \mathcal{R}) < +\infty \text{ from } x\}}$.

A.6 Repair pseudometric. $d_R(d_i, d_j) = \inf_{\gamma: d_i \rightsquigarrow d_j} \int_{\gamma} c \, ds$, defining a metric on $\text{Rest}(\mathcal{R})$ in the reversible case.

A.7 Repair Boundary Theorem (*proved in [1]*). $x \in \partial A(\mathcal{R})$ if and only if $C_R(d, \mathcal{R}) \rightarrow +\infty$ for some $d \in \text{Rest}(\mathcal{R})$ as the system approaches x .

A.8 Repair Monotonicity Theorem (*proved in [1]*). $\mathcal{R}_1 \subseteq \mathcal{R}_2$ implies $\text{Rest}(\mathcal{R}_1) \subseteq \text{Rest}(\mathcal{R}_2)$.

A.9 Paradigm Shift Theorem (*proved in [1]*). A non-trivial repair-repertoire change produces $\text{Rest}(\mathcal{R}_{\text{old}}) \triangle \text{Rest}(\mathcal{R}_{\text{new}}) \neq \emptyset$.

B. The Evaluability Measure

The main text treats evaluability as binary: a distinction d is either evaluable ($d \in \text{Rest}(\mathcal{R})$) or unevaluable ($d \notin \text{Rest}(\mathcal{R})$). This appendix develops a graded version in which evaluability is a real-valued measure on the distinction space, vanishing at the irrestorability boundary and positive in the interior.

Definition B.1 (Evaluability measure). Let $f : [0, +\infty] \rightarrow [0, 1]$ be a continuous, strictly decreasing function with $f(0) = 1$ and $\lim_{c \rightarrow \infty} f(c) = 0$. The *evaluability measure* of distinction d under repertoire $\mathcal{R}$ is

$$\mu_{\mathcal{R}}(d) = f(C_R(d, \mathcal{R})) \in [0, 1].$$

Natural choices include $f(c) = e^{-c}$ (exponential decay) and $f(c) = (1 + c)^{-1}$ (hyperbolic decay).

The evaluability measure extends the binary domain restriction to a soft version: $\mu_{\mathcal{R}}(d) = 1$ for costlessly restorable distinctions, $\mu_{\mathcal{R}}(d) \approx 0$ for near-irrestorable distinctions, and $\mu_{\mathcal{R}}(d) = 0$ precisely when $d \notin \text{Rest}(\mathcal{R})$. The binary Domain Restriction Theorem is the $\{0, 1\}$ -valued special case obtained by thresholding at any level $\epsilon > 0$.

Proposition B.2 (Monotonicity of the evaluability measure). *If $\mathcal{R}_1 \subseteq \mathcal{R}_2$, then $\mu_{\mathcal{R}_1}(d) \leq \mu_{\mathcal{R}_2}(d)$ for all $d \in \mathcal{D}$. Acquiring more repair operators increases evaluability pointwise.*

Proof. By Repair Monotonicity [1], $C_R(d, \mathcal{R}_2) \leq C_R(d, \mathcal{R}_1)$ for all d : a larger repertoire can only reduce or maintain minimal repair cost. Since f is strictly decreasing, $f(C_R(d, \mathcal{R}_2)) \geq f(C_R(d, \mathcal{R}_1))$, giving $\mu_{\mathcal{R}_2}(d) \geq \mu_{\mathcal{R}_1}(d)$. $\square$

Proposition B.3 (Boundary behaviour). $\mu_{\mathcal{R}}(d) \rightarrow 0$ as d approaches the irreversibility boundary $\partial A(\mathcal{R})$ in the repair pseudometric.

Proof. By the Repair Boundary Theorem [1], $C_{\mathcal{R}}(d, \mathcal{R}) \rightarrow +\infty$ as $d \rightarrow \partial A(\mathcal{R})$. Since $f(c) \rightarrow 0$ as $c \rightarrow +\infty$, $\mu_{\mathcal{R}}(d) = f(C_{\mathcal{R}}(d, \mathcal{R})) \rightarrow 0$. $\square$

Definition B.4 (Total evaluability). Let ν be a reference measure on $\mathcal{D}$. The *total evaluability* of observer $(S, \mathcal{R})$ is

$$E(\mathcal{R}) = \int_{\mathcal{D}} \mu_{\mathcal{R}}(d) \, d\nu(d) = \int_{\mathcal{D}} f(C_{\mathcal{R}}(d, \mathcal{R})) \, d\nu(d).$$

Proposition B.5 (Total evaluability and repair-repertoire reduction). Let $(\mathcal{R}_t)_{t \geq 0}$ be an erasure path in the sense of [2]. Then $E(\mathcal{R}_t)$ is non-increasing in t : progressive erasure weakly reduces total evaluability.

Proof. For each d , $C_{\mathcal{R}}(d, \mathcal{R}_t)$ is non-decreasing in t along the erasure path (by the monotonicity of hysteresis under progressive erasure, proved in [2]). Since f is decreasing, $\mu_{\mathcal{R}_t}(d)$ is non-increasing in t for each d . By dominated convergence (each integrand is bounded in $[0, 1]$), $E(\mathcal{R}_t) = \int \mu_{\mathcal{R}_t} \, d\nu$ is non-increasing. $\square$

Remark B.6. Proposition B.5 connects the evaluability measure to the erasure geometry of [2]: the total evaluability $E(\mathcal{R}_t)$ is the integral of the evaluability measure over the distinction space, and it decreases monotonically as the admissibility manifold deforms under erasure. The rate $-dE/dt$ is a scalar measure of how fast the observer’s domain of truth is shrinking — a global version of the boundary migration measure defined in [2]. When $f = e^{-c}$, total evaluability $E(\mathcal{R})$ is the Laplace transform of the repair cost distribution, evaluated at $s = 1$, giving a direct connection to the moment-generating function of $C_{\mathcal{R}}$.

C. Domain Restriction in Continuous Resolution Towers

Section 5 showed that SAHC’s finite hierarchy $h \in \{1, \dots, H\}$ instantiates the Domain Restriction Theorem at each level. This appendix extends the Hierarchy Consistency Corollary (Corollary 5.1) to continuous resolution towers, which is the setting required for the full TARTAN framework.

Definition C.1 (Resolution tower). A *resolution tower* is a directed family of repair repertoires $(\mathcal{R}_s)_{s \in [0, \infty)}$ indexed by a continuous resolution parameter s ,

with $\mathcal{R}_s \subseteq \mathcal{R}_{s'}$ whenever $s \leq s'$. Higher s corresponds to finer resolution. The associated restorable distinction spaces form a nested family: $\text{Rest}(\mathcal{R}_s) \subseteq \text{Rest}(\mathcal{R}_{s'})$ for $s \leq s'$, by Repair Monotonicity.

Definition C.2 (Transport operator). A *transport operator* between resolutions $s \leq s'$ is a map $T_{s \rightarrow s'} : \text{Rest}(\mathcal{R}_s) \rightarrow \text{Rest}(\mathcal{R}_{s'})$ satisfying:

1. $T_{s \rightarrow s} = \text{id}$ (identity at equal resolutions);
2. $T_{s \rightarrow s''} = T_{s' \rightarrow s''} \circ T_{s \rightarrow s'}$ for all $s \leq s' \leq s''$ (cocycle condition);
3. $T_{s \rightarrow s'}$ is admissibility-preserving: if $\mathcal{J}_s(d) \in \{0, 1\}$ then $\mathcal{J}_{s'}(T_{s \rightarrow s'}(d)) \in \{0, 1\}$.

The cocycle condition makes the family $(T_{s \rightarrow s'})$ into a continuous analogue of the composition $\iota_{H, H-1} \circ \dots \circ \iota_{2, 1}$ from the finite hierarchy case. Condition (3) is the continuous version of admissibility-preservation.

Theorem C.3 (Continuous Hierarchy Consistency Theorem). *Let $(\mathcal{R}_s)_{s \in [0, \infty)}$ be a resolution tower with transport operators $(T_{s \rightarrow s'})$. Suppose a family of truth predicates $(\mathcal{J}_s)_{s \geq 0}$ satisfies*

$$\mathcal{J}_s(d) = \mathcal{J}_{s'}(T_{s \rightarrow s'}(d)) \quad \text{for all } s \leq s' \text{ and all } d \in \text{Rest}(\mathcal{R}_s).$$

Then each $\mathcal{J}_s$ satisfies the Domain Restriction Theorem: $\text{dom}(\mathcal{J}_s) \subseteq \text{Rest}(\mathcal{R}_s)$. Moreover, the assignment $s \mapsto \text{dom}(\mathcal{J}_s)$ is monotone: coarser resolutions have smaller evaluability domains.

Proof. Fix $s \geq 0$. Suppose $d \in \text{dom}(\mathcal{J}_s)$, so $\mathcal{J}_s(d) \in \{0, 1\}$. We must show $d \in \text{Rest}(\mathcal{R}_s)$.

By Lemma 3.3 applied at resolution s , truth-evaluation of d at resolution s requires the existence of a finite-cost repair path under $\mathcal{R}_s$. Therefore $C_R(d, \mathcal{R}_s) < +\infty$, so $d \in \text{Rest}(\mathcal{R}_s)$.

Monotonicity: for $s \leq s'$, if $d \in \text{dom}(\mathcal{J}_s)$ then $d \in \text{Rest}(\mathcal{R}_s) \subseteq \text{Rest}(\mathcal{R}_{s'})$, so $T_{s \rightarrow s'}(d) \in \text{Rest}(\mathcal{R}_{s'})$ and $\mathcal{J}_{s'}(T_{s \rightarrow s'}(d)) = \mathcal{J}_s(d) \in \{0, 1\}$, giving $T_{s \rightarrow s'}(d) \in \text{dom}(\mathcal{J}_{s'})$. Therefore the evaluability domain grows with resolution: $T_{s \rightarrow s'}(\text{dom}(\mathcal{J}_s)) \subseteq \text{dom}(\mathcal{J}_{s'})$. $\square$

Definition C.4 (TARTAN coherence functional). Given a resolution tower with transport operators, the *TARTAN coherence functional* at resolutions $s_1 < s_2 < \dots < s_n$ is

$$L_{\text{TARTAN}} = \sum_{i < j} d_R(T_{s_i \rightarrow s_j}(x_{s_i}), x_{s_j}),$$

where d_R is the repair pseudometric and x_s denotes the observer’s distinction-space representation at resolution s .

Proposition C.5 (Zero coherence implies consistent domain restriction). *If $L_{\text{TARTAN}} = 0$ for all pairs (s_i, s_j) , then the truth predicates $(\mathcal{J}_s)$ satisfy the consistency condition of Theorem C.3 and the Domain Restriction Theorem holds at every resolution.*

Proof. $L_{\text{TARTAN}} = 0$ implies $T_{s_i \rightarrow s_j}(x_{s_i}) = x_{s_j}$ in the repair pseudometric for all pairs, which is exactly the consistency condition $\mathcal{J}_{s_i} = \mathcal{J}_{s_j} \circ T_{s_i \rightarrow s_j}$. Theorem C.3 then applies. $\square$

Remark C.6. Proposition C.5 gives a precise sense in which L_{TARTAN} is a penalty for violating the Domain Restriction Theorem across the continuous resolution tower. Minimising L_{TARTAN} during training drives the system toward a state where truth assignments at every resolution are mutually consistent and each is restricted to its own admissibility domain. SAHC’s consensus loss L_{HC} is the finite, discrete-hierarchy instance of L_{TARTAN} : it penalises the same violation at finitely many resolution levels using Jensen–Shannon divergence as the specific choice of d_R . The continuous version in Definition C.4 applies equally to spatial resolution towers (pixels at 10 m, 100 m, 1 km), temporal resolution towers (daily, monthly, annual), and semantic resolution towers (species, genus, family), and to systems where all three hierarchies are active simultaneously.

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